ZAMBIA UNIVERSITY COLLEGE OF TECHNOLOGY

ICT 2205 Data Structures and Algorithms

Final examination - Academic year 2020, semester 1

Time allowed: 3 hours Total marks: 100 Lecturer: Mrs J. Chibwe

Instructions: answer ALL questions. Write clearly. Show all working.

Question 1: Complexity

(a) The time complexity of binary search on a sorted array is:

(b) Which grows fastest as n increases?

(c) Big-O notation describes an algorithm's:

(d) Accessing an array element by index is:

Question 2: Linked lists

(a) Inserting at the head of a singly linked list takes:

(b) Each node of a doubly linked list stores:

(c) Finding the k-th element of a linked list takes:

(d) A circular linked list's last node points to:

Question 3: Sorting

(a) Which sort has worst case O(n^2) but average O(n log n)?

(b) Which sorting algorithm is stable and always O(n log n)?

(c) Bubble sort's best case on an already sorted list (with early exit) is:

(d) Which sort repeatedly selects the minimum and places it at the front?